Performance Testing

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask test client (functional checks) + Playwright (browser-driven checks). No dedicated load-testing tool (e.g. JMeter/Locust/k6) was used in this build.
Type of Testing	Functional response-time testing under single-user conditions; not formal load/stress/spike testing
Target Module / API	/predict, /api/predict, /batch/predict
Test Environment	Local development server (Flask dev server)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single-applicant prediction round trip	1	< 1	Decision returned well under 1 second
2	Batch CSV scoring (8 applicants)	1	< 1	All 8 rows scored without errors
3	Full model training (4 algorithms on ~6,400 rows)	1	< 10	All 4 models trained and compared without errors
4	Concurrent multi-user load	Not tested in this build	—	Not assessed — see Observations below

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	< 1 second	Pass	Model loaded once at app startup; inference itself is near-instant
2	Response Time (Max)	< 5 seconds	< 1 second	Pass	No case observed exceeding 1 second locally
3	Throughput (Req/sec)	—	Not measured	Not tested	No load-testing tool was used to measure sustained throughput
4	Error Rate	< 1%	0% (across all manual + scripted tests)	Pass	No errors across single, batch, and API test runs
5	CPU Utilization	< 80%	Not measured	Not tested	Not profiled in this build
6	Memory Utilization	< 80%	Not measured	Not tested	Not profiled in this build

Step 4: Observations & Analysis

Key Findings:
The ML model returns Approve/Decline decisions in under 1 second for single applicants and under 1 second for a batch of 8. All 8 functional and performance test cases passed with 0% error rate. Strong applicant profiles (high income, long employment) correctly received high approval confidence (99.5%), while risky profiles (low income, student, no employment) received high decline risk (93.6%).

Bottlenecks Identified:
No bottlenecks observed under single-user testing. Concurrent multi-user load was not tested in this build — this remains an open item for production deployment.

Optimization Steps Taken:
During development, the initial model used class-balanced training which caused borderline-good applicants to be over-rejected. This was fixed by removing class balancing and calibrating the decision threshold using F1-score maximization on held-out data, which significantly improved fairness of the decisions.

Step 5: Screenshots / Evidence

Ledgerline
CREDIT DECISIONING

- New application
- Batch screening
- Model report

● Model live
Engine: Logistic Regression

UNDERWRITING RESULT

Application decision

APPROVED

Approval confidence	73.9%
Decline risk	26.1%
Model engine	Logistic Regression
Annual income on file	$95,000
Years employed	8
Income type	Working

← New application

Ledgerline
CREDIT DECISIONING

New application
Batch screening
Model report

● Model live
Engine: Logistic Regression

COMPLIANCE REVIEW

Batch applicant screening

Upload a CSV of applicants to screen them all at once. Each row is run through the same model used for single applications, so high-risk applicants are flagged consistently at any volume.

Choose File No file chosen

CSV columns: gender, age, family_status, fam_members, children, education_type, income, income_type, years_employed, occupation_type, own_realty, own_car, housing_type, work_phone, phone, email

Run batch screen →

8 SCREENED **4** APPROVED **4** DECLINED

INCOME	INCOME TYPE	YEARS EMPLOYED	FAMILY STATUS	DECISION	DECLINE RISK
$95,000	Working	8	Married	APPROVED	31.0%
$250,000	Working	20	Married	APPROVED	0.3%
$27,000	Student	0	Single / not married	DECLINED	93.6%
$45,000	Working	2	Single / not married	DECLINED	64.0%
$68,000	Pensioner	0	Widow	APPROVED	10.9%
$38,000	Commercial associate	1	Civil marriage	DECLINED	85.5%
$180,000	State servant	30	Married	APPROVED	0.7%
$22,000	Student	0	Single / not married	DECLINED	91.5%

Download full results (CSV) ↓